# Sina Masoumi Shahrbabak


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## Summary


PhD researcher developing machine learning and signal processing methods for physiological time-series and multimodal health data. Experienced in deep learning, Bayesian inference, and cardiovascular modeling using ECG, PPG, SCG, and blood-pressure waveforms. Published work spans wearable monitoring, medical diagnostics, and clinical data analysis, with advanced proficiency in Python and PyTorch.


## Education

**Doctor of Philosophy (PhD), Mechanical Engineering** | University of Maryland — *Expected Dec 2027*
Monitoring, diagnosis, and modelling of cardiovascular disease

**Master of Science, Mechanical Engineering** | University of Maryland — *May 2025*
Notable Courses: Computational Statistics, Machine Learning: Theory and Applications, Applied Machine Learning for Engineering, Optimal Estimation of Dynamical Systems — GPA: 3.9

**Bachelor of Science, Mechanical Engineering** | Sharif University of Technology — *2022*
GPA: 3.9

## Technical Skills

**Programming Languages:** Python, R, MATLAB
**Deep Learning & ML Frameworks:** PyTorch, Optuna, Scikit-learn
**Methodologies & Algorithms:** Signal Processing, Time-Series Analysis, CNNs, RNNs, LSTMs, GRUs, Bayesian Inference & MCMC
**Dynamical Systems & Control:** System Identification, State Estimation, Kalman Filtering, Nonlinear Control

## Research Experience

**Graduate Research Assistant, University of Maryland | College Park, MD** — *2023 to Present*

**Wearable Hemodynamic Monitoring**
- Collected multimodal physiological data from 20 swine and developed signal-processing and machine-learning pipelines for more than 300,000 cardiac cycles.
- Developed wearable stroke volume estimation models using ECG, PPG, and SCG, achieving a median absolute percentage error of 8.8% during hemorrhage and blood transfusion.

**Vascular Disease Detection**
- Designed CNN models to detect abdominal aortic aneurysm and peripheral artery disease from blood pressure waveforms.

**Cardiovascular Mathematical Modeling**
- Developed mathematical and computational models of cardiovascular dynamics, arterial pressure propagation, and ballistocardiography.

**Research Intern, Massachusetts General Hospital | Boston, MA** — *Summer 2025*

**Mathematical Modeling of Vasoactive Medication Dose Response**
- Developed a Bayesian inference framework to correct medication dose-change timing errors in retrospective ICU data.
- Analyzed continuous blood pressure and vasoactive medication records to characterize patient-specific dose-response relationships.

## Mentorship

**University of Maryland, College Park** — *Spring 2026*
Mentored an undergraduate research intern and a high school student conducting projects on ballistocardiogram simulation during cold-water immersion and pulse-transit-time-based blood pressure monitoring.

## Selected Publications (5 of 11)


- **Masoumi Shahrbabak, S.**, Zhou, Y., Bahrami, R., Jafari Ozoumchelooei, H., Tangolar, D., Rezaei, P., Kim, N., Vandenberge, J., Perez, R., Burch, N., Kim, D., Johnson, C. A., Jr., Tran, D., Vetri, R., Kimball, J. P., Mukkamala, R., Wu, Z. J., Inan, O. T., & Hahn, J. O. (2026). Benchmarking wearable physio-markers for stroke volume tracking during hemorrhage and blood transfusion in a swine model. *Under review.*
- **Masoumi Shahrbabak, S.**, Kim, S., Youn, B. D., Cheng, H. M., Chen, C. H., Mukkamala, R., & Hahn, J. O. (2024). Peripheral artery disease diagnosis based on deep learning-enabled analysis of non-invasive arterial pulse waveforms. *Computers in Biology and Medicine.*

- **Masoumi Shahrbabak, S.**, Youn, B. D., Cheng, H. M., Chen, C. H., Sung, S. H., Mukkamala, R., & Hahn, J. O. (2025). Deep learning-enabled diagnosis of abdominal aortic aneurysm using pulse volume recording waveforms: an in-silico study. *Sensors*.
- **Masoumi Shahrbabak, S.**, Mousavi, A., Mukkamala, R., & Hahn, J. O. (2025). In silico investigation of a mathematical model relating the ballistocardiogram to aortic blood pressure. *IEEE Transactions on Biomedical Engineering*.
- Zhou, Y., **Masoumi Shahrbabak, S.**, Rezaei, P., Bahrami, R., Rahman, F. N., Sanchez-Perez, J. A., Gazi, A. H., Inan, O. T., & Hahn, J. O. (2026). A virtual experiment generator to replicate cardiovascular responses to acute mental stress and transcutaneous median nerve stimulation. *Journal of Dynamic Systems*, *Measurement*, *and Control*.

## Selected Presentations

**Workshop Speaker,** "Digital Health Technologies and AI for Studying and Predicting Episodic Health Events," IEEE-EMBS International Conference on Biomedical and Health Informatics, 2025.

## Honors & Awards

IEEE BHI NSF-EMBS-Google Young Professional NextGen Scholar, 2025